

A Technical Introduction to Image segmentation

Section 1

Fast marching methods The fast marching method has been used in image segmentation, and this model has been improved (permitting both positive and negative propagation speeds) in an approach called the generalized fast marching method.

Section 2

Image segmentation using MAP and expectation–maximization The expectation–maximization algorithm is utilized to iteratively estimate the a posteriori probabilities and distributions of labeling when no training data is available and no estimate of segmentation model can be formed. A general approach is to use histograms to represent the features of an image and proceed as outlined briefly in this three-step algorithm: 1. A random estimate of the model parameters is utilized. 2. E step: Estimate class statistics based on the random segmentation model defined. Using these, compute the conditional probability of belonging to a label given the feature set is calculated using naive Bayes' theorem.

Section 3

In digital image processing and computer vision, image segmentation is the process of partitioning a digital image into multiple image segments, also known as image regions or image objects (sets of pixels). The goal of segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze. Image segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in images. More precisely, image segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics. The result of image segmentation is a set of segments that collectively cover the entire image, or a set of contours extracted from the image (see edge detection). Each of the pixels in a region are similar with respect to some characteristic or computed property, such as color, intensity, or texture. Adjacent regions are significantly different with respect to the same characteristic(s). When applied to a stack of images, typical in medical imaging, the resulting contours after image segmentation can be used to create 3D reconstructions with the help of geometry reconstruction algorithms like marching cubes.

Section 4

Parametric methods Lagrangian techniques are based on parameterizing the contour according to some sampling strategy and then evolving each element according to

image and internal terms. Such techniques are fast and efficient, however the original "purely parametric" formulation (due to Kass, Witkin and Terzopoulos in 1987 and known as "snakes"), is generally criticized for its limitations regarding the choice of sampling strategy, the internal geometric properties of the curve, topology changes (curve splitting and merging), addressing problems in higher dimensions, etc.. Nowadays, efficient "discretized" formulations have been developed to address these limitations while maintaining high efficiency. In both cases, energy minimization is generally conducted using a steepest-gradient descent, whereby derivatives are computed using, e.g., finite differences.

Section 5

Markov random fields The application of Markov random fields (MRF) for images was suggested in early 1984 by Geman and Geman. Their strong mathematical foundation and ability to provide a global optimum even when defined on local features proved to be the foundation for novel research in the domain of image analysis, de-noising and segmentation. MRFs are completely characterized by their prior probability distributions, marginal probability distributions, cliques, smoothing constraint as well as criterion for updating values. The criterion for image segmentation using MRFs is restated as finding the labelling scheme which has maximum probability for a given set of features. The broad categories of image segmentation using MRFs are supervised and unsupervised segmentation.

Section 6

the algorithm selects the newly formed graph. Simulated annealing requires the input of temperature schedules which directly affects the speed of convergence of the system, as well as energy threshold for minimization to occur.

Section 7

Here $\lambda \in \Lambda$ $\{\displaystyle \lambda \in \Lambda\}$, the set of all possible labels. 3. M step: The established relevance of a given feature set to a labeling scheme is now used to compute the a priori estimate of a given label in the second part of the algorithm. Since the actual number of total labels is unknown (from a training data set), a hidden estimate of the number of labels given by the user is utilized in computations.

Section 8

The simplest method of image segmentation is called the thresholding method. This method is based on a clip-level (or a threshold value) to turn a gray-scale image into a binary image. The key of this method is to select the threshold value (or values when multiple-levels are selected). Several popular methods are used in industry

including the maximum entropy method, balanced histogram thresholding, Otsu's method (maximum variance), and k-means clustering. Recently, methods have been developed for thresholding computed tomography (CT) images. The key idea is that, unlike Otsu's method, the thresholds are derived from the radiographs instead of the (reconstructed) image. New methods suggest the use of multi-dimensional, fuzzy rule-based, non-linear thresholds. In these approaches, the decision regarding each pixel's membership in a segment is based on multi-dimensional rules derived from fuzzy logic and evolutionary algorithms, considering factors such as image lighting, environment, and application.

Section 9

Image segmentation and primal sketch There have been numerous research works in this area, out of which a few have now reached a state where they can be applied either with interactive manual intervention (usually with application to medical imaging) or fully automatically. The following is a brief overview of some of the main research ideas that current approaches are based upon. The nesting structure that Witkin described is, however, specific for one-dimensional signals and does not trivially transfer to higher-dimensional images. Nevertheless, this general idea has inspired several other authors to investigate coarse-to-fine schemes for image segmentation. Koenderink proposed to study how iso-intensity contours evolve over scales and this approach was investigated in more detail by Lifshitz and Pizer. Unfortunately, however, the intensity of image features changes over scales, which implies that it is hard to trace coarse-scale image features to finer scales using iso-intensity information. Lindeberg studied the problem of linking local extrema and saddle points over scales, and proposed an image representation called the scale-space primal sketch which makes explicit the relations between structures at different scales, and also makes explicit which image features are stable over large ranges of scale including locally appropriate scales for those. Bergholm proposed to detect edges at coarse scales in scale-space and then trace them back to finer scales with manual choice of both the coarse detection scale and the fine localization scale. Gauch and Pizer studied the complementary problem of ridges and valleys at multiple scales and developed a tool for interactive image segmentation based on multi-scale watersheds. The use of multi-scale watershed with application to the gradient map has also been investigated by Olsen and Nielsen and been carried over to clinical use by Dam. Vincken et al. proposed a hyperstack for defining probabilistic relations between image structures at different scales. The use of stable image structures over scales has been furthered by Ahuja and his co-workers into a fully automated system. A fully automatic brain segmentation algorithm based on closely related ideas of multi-scale watersheds has been presented by Undeman and

Lindeberg and been extensively tested in brain databases. These ideas for multi-scale image segmentation by linking image structures over scales have also been picked up by Florack and Kuijper. Bijaoui and Rué associate structures detected in scale-space above a minimum noise threshold into an object tree which spans multiple scales and corresponds to a kind of feature in the original signal. Extracted features are accurately reconstructed using an iterative conjugate gradient matrix method.